



University of Rajshahi
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B.Sc. Engineering Part IV Even Semester Exam-2024
Title: Distributed Database Management System Lab

Question 1: (a) Create a database named **HospitalDB**. Under HospitalDB, create the following three tables. The **Doctor** table is not partitioned. **Patient** and **Visit** tables must be partitioned by **department_id**.

i. Doctor Table (Not Partitioned)

Column	Description
doctor_id	BIGINT
name	VARCHAR(100)
specialization	VARCHAR(100)
email	VARCHAR(100)

ii. Patient Table (Partitioned by department_id)

Column	Description
patient_id	BIGINT
name	VARCHAR(100)
age	INT
department_id	INT

iii. Visit Table (Partitioned by department_id)

Column	Description
visit_id	BIGINT
patient_id	BIGINT
department_id	INT
fee	DECIMAL(10,2)

(b) Insert **Doctor.csv**, **Patient.csv**, and **Visit.csv** data into respective tables. Use temporary tables if required, and delete them after insertion.

- Doctor: 150 records → **Doctor.csv**
- Patient: 200 records → **Patient.csv**
- Visit: 500 records → **Visit.csv**

(c) Write **Hive commands** to add a new partition (**department_id=7**) to **Patient** and **Visit** tables.

(d) Write **Hive commands** to drop partition (**department_id=4**) from **Patient** and **Visit** tables.

(e) Display the total number of patient visits in each partition.

Question 2: Write a MapReduce program to find the top-3 most frequent words in a large dataset.

Mapper: For each word, emit (**word, 1**).

Reducer: For each word, emit (**word, aggregate counts**) and output top-2 words.

Sample Input: *mango apple banana apple orange banana apple orange banana mango banana orange*

Mapper Output:	Reducer Output:
mango, 1)	(mango, 2)
(apple, 1)	(apple, 3)
(banana, 1)	(banana, 4)
(apple, 1)	(orange, 3)
(orange, 1)	
(banana, 1)	
(apple, 1)	
(orange, 1)	
(banana, 1)	
(mango, 1)	
(banana, 1)	
(orange, 1)	